

Topology of 3-Manifolds Supporting Nonsingular Morse–Smale Flows with One Saddle Orbit

O. Pochinka, D. Shubin, A. Yagilev

HSE University

Why study Morse–Smale flows?

- Structurally stable dynamics
- Finite number of periodic orbits
- Global topology strongly constrained by dynamics

Dynamics $\longrightarrow$ Topology

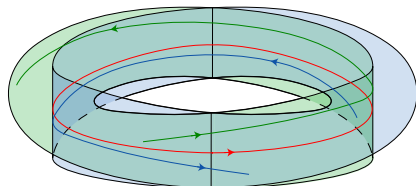
The guiding question

What can a single saddle orbit force globally?

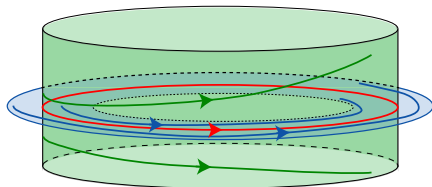
- **Morgan:** nonsingular Morse–Smale flows exist only on graph manifolds.
- **Two periodic orbits:** only lens spaces appear.
- **Three periodic orbits:** small Seifert manifolds appear.
- **Twisted saddle orbit:** classification obtained in our previous work.

Today: the non-twisted saddle case.

Twisted Saddle



Untwisted saddle

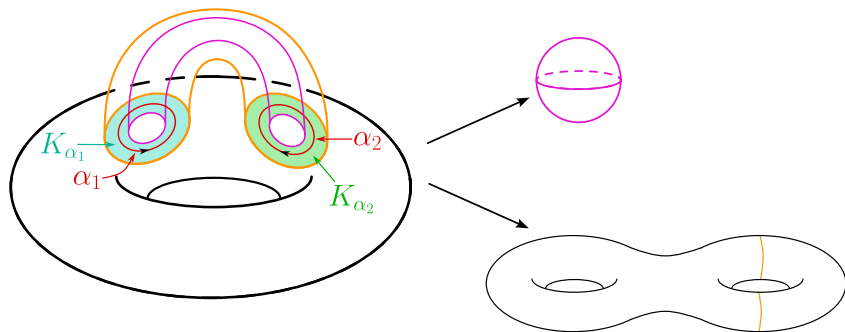


Each periodic orbit has a canonical solid torus neighborhood.

The saddle handle attaches through two annuli.

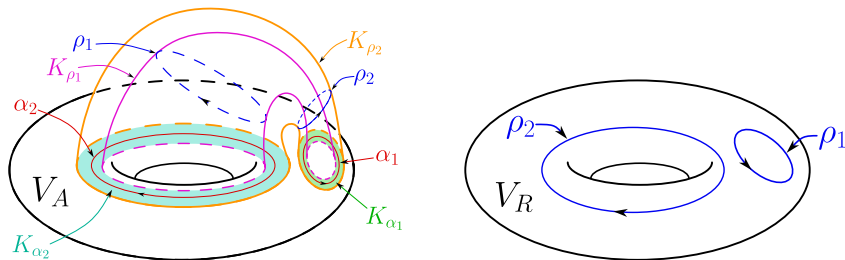
The key geometric restriction

The boundary must remain a union of tori.



Higher genus boundary appears $\Rightarrow$ impossible.

Only a few attachment patterns survive



One saddle orbit rigidifies the gluing combinatorics.

Theorem

A connected closed orientable 3-manifold admits a nonsingular Morse–Smale flow with exactly one saddle orbit if and only if it is homeomorphic to

- *a lens space,*
- *a connected sum of lens spaces,*
- *or a Seifert manifold*

$$M(S^2, (p_1, q_1), (p_2, q_2), (p_3, q_3)).$$

What produces these manifolds?

Lens spaces

Seifert manifolds

Different surviving gluings produce different ambient topologies.

A single saddle orbit
rigidifies the ambient topology.

The restriction comes from torus boundary combinatorics.

Future directions

- Classification of topological equivalence classes.
- Relation between dynamics and JSJ-type topology.
- Flows with several saddle orbits.
- Dynamical invariants vs topology of ambient manifolds.

Topology from dynamics.

Thank you!

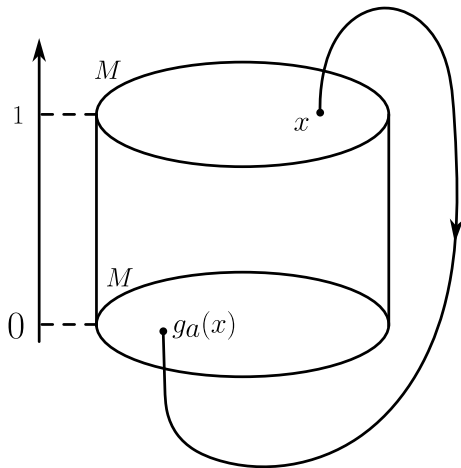
Questions?

Definition of Morse–Smale flow

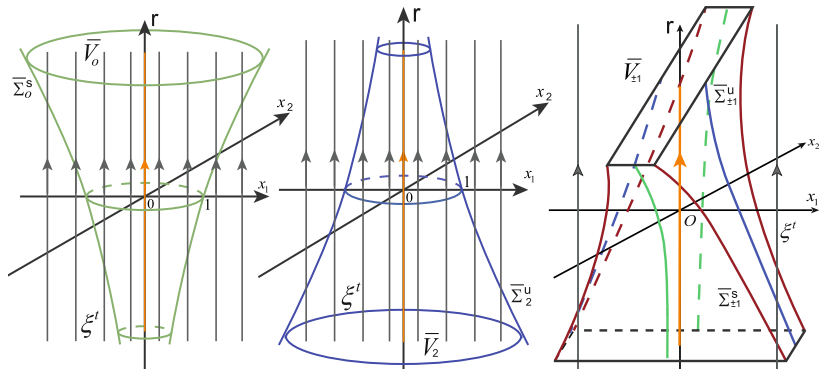
A nonsingular flow f^t is Morse–Smale if:

- 1 the nonwandering set consists of finitely many hyperbolic periodic orbits;
- 2 stable and unstable manifolds intersect transversally.

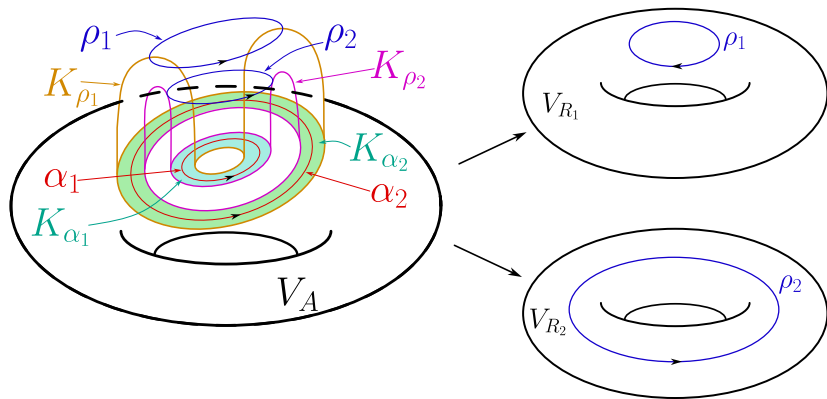
Suspension model



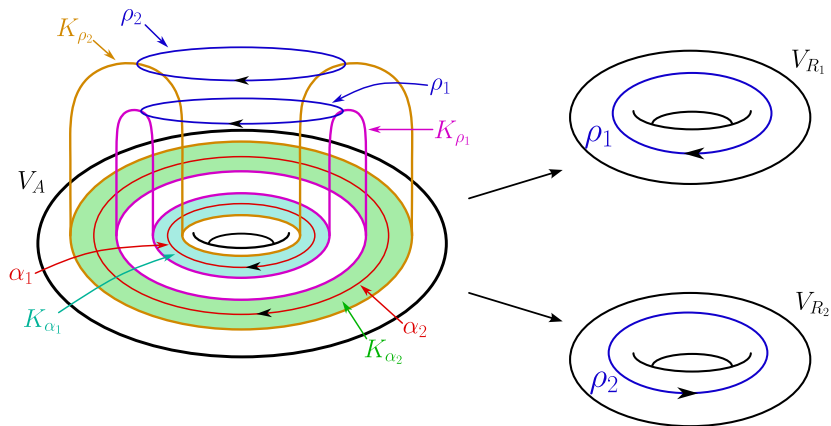
Canonical neighborhoods



Additional attachment cases



Additional attachment cases



Additional attachment cases

